Economic Paper Reproduction

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Abstract

The paper *Vouchers for Private Schooling in Colombia: Evidence from a Randomized Natural Experiment* by Joshua Angrist, Eric Bettinger, Erik Bloom, Elizabeth King, and Michael Kremer examines the effects of Colombia's PACES program, which distributed more than 125,000 vouchers to low-income students to help cover the cost of private secondary schooling. Because vouchers were awarded by lottery, the program provides a natural experiment for strong causal inference. The authors compare lottery winners and losers, using reduced-form models and instrumental variable (IV) techniques to estimate the impact of private school access on a range of educational and non-educational outcomes.

The study finds that receiving a PACES voucher significantly improved educational outcomes, including increasing attendance at private schools, reducing grade repetition, increasing test scores, and extending the time of teen marriage and employment. In this project, we replicate several key empirical results, including summary statistics (Table 1), comparisons of personal characteristics and voucher status (Table 2), and impacts on educational outcomes and costs (Tables 4, 5, and 6), to better understand the strengths of randomized evaluations and evaluate the scalability of voucher-based interventions in low-income settings.

1 Introduction

The use of school vouchers as a tool to enhance educational success in developing nations has drawn greater public interest in recent decades. Although there has been much discussion about voucher programs in the US, their applicability may be even more significant in low- and middle-income nations where private school enrollment rates are relatively higher and public school quality is frequently significantly worse. One of the most thorough empirical analyses of a large-scale voucher program in a developing nation can be found in the study Vouchers for Private Schooling in Colombia: Evidence from a Randomized Natural Experiment by Angrist, Bettinger, Bloom, King, and Kremer (2002). This study uses the random assignment of vouchers to estimate causal impacts.

More than 125,000 vouchers were given to low-income students in Colombia as part of the PACES initiative to assist with the expense of private secondary education. In overcrowded cities, vouchers were given out by lottery, creating an ideal environment for the experiment. The authors compare lottery winners (treatment group) to lottery losers (control group) using data from three cohorts of applicants in Bogotá and Jamundí between 1993 and 1997. Through the use of instrumental variable (IV) approaches and reduced-form comparisons, they discover that receiving a voucher resulted in notable improvements in educational outcomes, including lower rates of teen marriage, higher test scores, higher private school attendance, and delayed labor market entry.

By reproducing some of the significant empirical findings from the original publication, the paper seeks to replicate and enhance comprehension of the PACES study. Table 1 presents summary statistics, while Table 2 compares baseline characteristics. We will process, clean, and analyze the original dataset using SQL, Python, and pandas techniques learned in earlier training. In addition to critically examining randomized experimental designs and their capacity to generate reliable estimates of program impacts, this replication exercise aims to improve our technical proficiency in empirical economics. In the end, this project

offers a chance to hone quantitative research abilities and consider the wider ramifications of demand-side finance interventions for educational policy in low-income settings.

2 Reproducibility

2.1 Tables 1 and 2

Table 1: Sample Design and Survey Response Data

Variable	Bogotá 1995	Bogotá 1997	Jamundí 1993	Combined sample	Test-takers
A. Population:					
N	4,044	1,770	343	6,157	–
Percentage awarded vouchers	58.8	84.7	50.1	65.8	–
B. Attempted Interviews:					
N	2,249	457	279	2,985	473
Percentage awarded vouchers	50.0	51.6	50.2	50.3	53.9
Response rate	0.523	0.606	0.591	0.542	0.598
Winner rate	0.528	0.619	0.650	0.553	0.624
Loser rate	0.518	0.593	0.532	0.531	0.571
C. Completed Interviews:					
N	1,176	277	165	1,618	283
Percentage awarded vouchers	50.4	52.7	55.2	51.3	55.6
Household visit	0.054	0.004	0.782	0.120	0.093
Age at time of application	12.6	12.4	12.5	12.6	12.6
(std dev)	(1.3)	(1.4)	(1.9)	(1.4)	(1.2)
Age on survey date	15.0	13.1	16.9	14.9	15.6
(std dev)	(1.3)	(1.4)	(1.5)	(1.7)	(1.2)
Male	0.511	0.505	0.424	0.494	0.523
Started 6th grade in private	0.910	0.880	0.669	0.880	0.833
Started 7th grade in private	0.763	0.731	0.626	0.744	0.731
Currently in private school	0.618	0.738	0.506	0.628	0.694
Highest grade completed	7.6	6.0	8.6	7.4	7.7
(std dev)	(0.94)	(0.48)	(1.10)	(1.10)	(0.91)
Currently in school	0.836	0.957	0.778	0.851	0.841

Using descriptive statistics, Table 1 summarizes survey response trends for Bogotá 1995, Bogotá 1997, and Jamundí 1993. We present counts (N), demographics (e.g., age, gender, and grade completion), percentages of voucher recipients, response rates, and attempted and completed interviews by population. Understanding the experimental design's initial structure requires a comprehension of this table, demonstrating that the lottery-based randomization produced equivalent treatment and control groups—a necessary condition for

reliable causal inference. Sorted summaries and pandas group-by operations were used to aggregate the data, guaranteeing precise fitting across several sample definitions. According to general econometrics, these summary statistics ensure that later regression models are based on well-balanced samples with low selection bias before calculating treatment effects.

Table 2: Personal Characteristics and Voucher Status

Dependent variable	Bogotá 1995		Bogotá 1997		Jamundí 1993		Combined sample		Test-takers	
	Loser mean	Won voucher	Loser mean	Won voucher	Loser mean	Won voucher	Loser mean	Won voucher	Loser mean	Won voucher
A. Data from PACES Application:										
Has phone	0.882	0.009	0.828	0.030	0.301	0.068	0.825	0.017	—	—
		(0.011)		(0.025)		(0.052)		(0.010)		
Age at time of application	12.7	-0.086	12.7	-0.228	12.7	-0.383	12.7	-0.133	—	—
	(1.327)	(0.045)	(1.531)	(0.122)	(1.504)	(0.162)	(1.371)	(0.040)		
Male	0.493	0.012	0.484	0.003	0.386	0.114	0.483	0.019	—	—
		(0.017)		(0.044)		(0.055)		(0.015)		
N	1519	3661	256	1736	166	334	1941	5731	—	—
B. Data for All Attempted Contacts:										
Has phone	1.000	0.000	1.000	0.000	0.370	0.082	0.938	0.008	—	—
		(0.000)		(0.004)		(0.059)		(0.006)		
Age at time of application	12.8	-0.119	12.6	-0.191	12.8	-0.595	12.7	-0.177	—	—
	(1.341)	(0.060)	(1.522)	(0.142)	(1.572)	(0.183)	(1.393)	(0.053)		
Male	0.500	-0.007	0.488	-0.020	0.372	0.102	0.486	0.001	—	—
		(0.022)		(0.048)		(0.061)		(0.019)		
N	1035	2067	212	448	135	272	1382	2787	—	—
C. Survey Data:										
Age at time of survey	15.0	-0.061	13.2	-0.288	17.2	-0.500	14.9	-0.145	14.9	-0.160
	(1.351)	(0.079)	(1.440)	(0.166)	(1.392)	(0.223)	(1.684)	(0.068)	(1.398)	(0.162)
Male	0.489	0.025	0.527	-0.040	0.365	0.108	0.484	0.022	0.516	0.012
		(0.029)		(0.060)		(0.077)		(0.025)		(0.060)
Mother's highest grade completed	5.9	-0.045	5.9	0.682	4.4	1.295	5.8	0.193	5.8	-0.192
	(2.669)	(0.171)	(2.747)	(0.357)	(2.717)	(0.481)	(2.714)	(0.146)	(2.617)	(0.325)
Father's highest grade completed	5.9	-0.361	5.5	1.035	5.2	0.717	5.8	-0.007	5.3	0.030
	(2.941)	(0.197)	(2.496)	(0.382)	(2.932)	(0.620)	(2.870)	(0.170)	(2.749)	(0.372)
Mother's age	40.7	0.030	38.7	-0.171	43.6	-1.146	40.6	-0.106	40.3	0.459
	(7.326)	(0.423)	(6.600)	(0.794)	(8.811)	(1.393)	(7.423)	(0.362)	(6.576)	(0.808)
Father's age	44.4	0.636	41.9	0.028	45.5	1.132	44.1	0.570	43.5	1.179
	(8.141)	(0.534)	(7.293)	(0.940)	(9.098)	(1.661)	(8.135)	(0.451)	(7.652)	(1.053)
Father's wage	0.100	0.009	0.088	0.002	0.133	-0.082	0.101	0.000	0.052	0.083
		(0.021)		(0.040)		(0.058)		(0.018)		(0.038)
N	583	1176	131	277	74	165	788	1618	124	283

By contrasting baseline traits and outcome variables between voucher winners and losers across three panels (application data, attempted contact data, and post-survey data) Table 2 furthers the descriptive analysis. We estimate mean differences in addition to standard deviations with coefficients using regression models with cohort fixed effects and city controls. This is a traditional method in applied econometrics to evaluate covariate balance. Furthermore, this table uses robust standard errors to account for clustering and incorporates both

continuous (e.g., age, parental education) and binary outcomes (e.g., possesses phone, male). The creation of this table required meticulous data cleaning, subsetting, and regression estimation. The outcomes were then converted into multi-index pandas DataFrames following the format of the original paper. Table 2 provides readers with econometric reassurance that the lottery win status is orthogonal to pre-treatment variables. This validates the use of the variable for causal identification in subsequent models that estimate the effects of vouchers on non-educational outcomes and educational attainment.

2.1.1 Challenges

For both tables, the regression results closely align with those reported in the original paper, with only minor variations in the coefficients, which fall within the acceptable range outlined by the course rubric (e.g., within ± 5 percentage of the original estimates).

2.2 Tables 3

Table 3 in Angrist et al. (2002) displays the reduced-form estimates of the impact of winning a PACES voucher lottery on various educational outcomes (including scholarship usage, school type, enrollment status, and grade progression) using the following specification:

$$y_{ic} = X_i'\beta_0 + \alpha_0 Z_i + \delta_c + \varepsilon_{ic} \tag{1}$$

These outcomes are reported for two samples: the Bogotá 1995 subset and the combined sample which includes the Bogotá 1995, Bogotá 1997, and Jamundí 1993 subsamples. Further, for each outcome, results are shown under three sets of controls: No controls, basic controls, and vasic + barrio controls (accounts for neighborhood fixed effects), and compared estimates between voucher winners and losers.

Regression estimation was conducted using OLS with robust standard errors. The models were implemented using statsmodels, after subsetting and cleaning the data using Pandas.

Table 3: Educational Outcomes and Voucher Status

	Bogotá 1995				Combined sample	
Dependent variable	Loser means (1)	No controls (2)	Basic controls (3)	Basic +19 barrio controls (4)	Basic controls (5)	Basic +19 barrio controls (6)
Using any scholarship in survey year	0.057	0.509	0.504	0.505	0.520	0.521
	(0.232)	(0.023)	(0.023)	(0.023)	(0.019)	(0.019)
Ever used a scholarship	0.244	0.672	0.663	0.662	0.636	0.635
	(0.430)	(0.021)	(0.022)	(0.022)	(0.019)	(0.019)
Started 6th grade in private	0.877	0.063	0.057	0.058	0.066	0.067
	(0.328)	(0.017)	(0.017)	(0.017)	(0.016)	(0.016)
Started 7th grade in private	0.673	0.174	0.168	0.171	0.170	0.173
	(0.470)	(0.025)	(0.025)	(0.025)	(0.022)	(0.021)
Currently in private school	0.539	0.160	0.153	0.156	0.152	0.154
	(0.499)	(0.028)	(0.028)	(0.028)	(0.023)	(0.023)
Highest grade completed	7.486	0.164	0.130	0.120	0.085	0.078
	(0.960)	(0.053)	(0.051)	(0.052)	(0.042)	(0.042)
Currently in school	0.831	0.019	0.007	0.007	−0.002	−0.002
	(0.375)	(0.022)	(0.020)	(0.020)	(0.016)	(0.016)
Finished 6th grade	0.943	0.026	0.023	0.021	0.014	0.012
	(0.232)	(0.012)	(0.012)	(0.012)	(0.011)	(0.011)
Finished 7th grade (excludes Bogotá 97)	0.847	0.040	0.031	0.029	0.027	0.025
	(0.360)	(0.020)	(0.020)	(0.020)	(0.018)	(0.018)
Finished 8th grade (excludes Bogotá 97)	0.632	0.112	0.100	0.094	0.081	0.077
	(0.483)	(0.027)	(0.027)	(0.027)	(0.025)	(0.025)
Repetitions of 6th grade	0.194	−0.066	−0.059	−0.059	−0.049	−0.049
	(0.454)	(0.024)	(0.024)	(0.025)	(0.020)	(0.020)
Ever repeated after lottery	0.224	−0.060	−0.055	−0.051	−0.055	−0.053
	(0.417)	(0.023)	(0.023)	(0.024)	(0.019)	(0.019)
Total repetitions since lottery	0.254	−0.073	−0.067	−0.064	−0.058	−0.057
	(0.508)	(0.028)	(0.028)	(0.028)	(0.022)	(0.022)
Years in school since lottery	3.653	0.058	0.034	0.031	0.015	0.012
	(0.951)	(0.053)	(0.051)	(0.051)	(0.044)	(0.044)
Sample size	562		1,147		1,577	

We created dummy variables for binary outcomes and categorical controls, and we one-hot encoded neighborhood controls.

Table 3 demonstrates that voucher winners were significantly more likely to use scholarships, attend private school, and complete more years of schooling. These findings are consistent across specifications and are robust to different sets of controls. These results provide strong evidence that lottery-based voucher assignment improves students' educational outcomes.

2.2.1 Challenges

The paper refers to a set of "basic controls" included in the regression models, but it fails to formally list them. We had to infer their composition from brief notes in the paper and from SAS script which was hard to decipher. Based on these clues from the paper, we inferred that the controls likely included variables such as age, gender, phone access, type of interview, socioeconomic strata, and month of interview.

Additionally, translating variable names in the dataset was not straightforward. The paper presented these names in English, but they were not directly labeled in the dataset using those terms. For example, it was not obvious that prscha_1 referred to "started private school in 7th grade". We had to reverse-engineer this connection from a combination of raw variables and logic found in the SAS code. The lack of direct mapping between variables and table labels required us to carefully check each regression definition and run our regressions based on assumptions that we correctly identified the variables.

Notably, the regression results closely align with those reported in the original paper, with minor variations falling within 5 percentage points of the paper's values.

2.3 Table 4

Table 4: Educational Outcomes and Voucher Status, By Gender

	Coefficient on voucher status					
	Bogotá 1995				Combined sample	
	Male		Female		Male	Female
Dependent variable	Loser means	Basic controls	Loser means	Basic controls	Basic controls	Basic controls
Started 6th grade in private	0.857	0.090	0.897	0.023	0.063	0.072
		(0.026)		(0.022)	(0.023)	(0.021)
Started 7th grade in private	0.646	0.192	0.699	0.144	0.169	0.176
		(0.036)		(0.034)	(0.031)	(0.030)
Currently in private school	0.543	0.136	0.535	0.171	0.124	0.182
		(0.039)		(0.039)	(0.033)	(0.033)
Highest grade completed	7.400	0.123	7.571	0.140	0.056	0.122
		(0.076)		(0.065)	(0.062)	(0.052)
Currently in school	0.843	−0.020	0.819	0.035	−0.026	0.029
		(0.029)		(0.027)	(0.024)	(0.022)
Finished 6th grade	0.932	0.014	0.954	0.032	0.003	0.027
		(0.018)		(0.013)	(0.017)	(0.012)
Finished 7th grade	0.825	0.026	0.869	0.041	0.019	0.039
		(0.029)		(0.025)	(0.027)	(0.022)
Finished 8th grade	0.589	0.095	0.674	0.105	0.071	0.090
		(0.039)		(0.036)	(0.037)	(0.032)
Repetitions of 6th grade	0.229	−0.087	0.160	−0.036	−0.070	−0.033
		(0.037)		(0.030)	(0.031)	(0.023)
Ever repeated after lottery	0.254	−0.083	0.195	−0.029	−0.076	−0.035
		(0.034)		(0.031)	(0.028)	(0.025)
Total repetitions since lottery	0.296	−0.101	0.213	−0.031	−0.079	−0.037
		(0.042)		(0.033)	(0.035)	(0.026)
Calendar years in school since lottery	3.664	−0.029	3.642	0.091	−0.041	0.081
		(0.077)		(0.063)	(0.067)	(0.055)
Sample size	280	575	282	572	779	798

Table 4 reports the reduced-form OLS estimates of winning a PACES voucher on a rich set of educational outcomes, separately for boys and girls in the 1995 Bogotá sample and in the full (Bogotá '95, Bogotá '97, Jamundí '93) combined cohort. In each panel the left-most column shows voucher losers' means, and the column labeled "Basic controls" gives the coefficient on the voucher-win indicator when controlling for city, year of application, phone ownership, age, survey type and instrument, residence strata, and month of interview

(with robust standard errors) . The dependent variables span private-school entry (starting 6th and 7th grades in private school; currently in private school), highest grade completed, current enrollment, completion of 6th–8th grades, grade-repetition measures (repetitions of 6th grade; ever repeated after the lottery; total repetitions since the lottery), and calendar years in school since the lottery.

Across both genders and samples, voucher winners are substantially more likely to enter private school – for instance, winning increases the probability of starting 6th grade in private school by about 8–19 percentage points – and complete more years of schooling, while also experiencing fewer grade repetitions. Separate results by sex show that although private-school take-up effects are similar for boys and girls, the gains in total years of schooling are somewhat larger and more precisely estimated for girls (0.14 years) than for boys (0.06 years) in the combined sample.

2.3.1 Challenges

In our Python+statsmodels replication of Table 4, most estimated coefficients align closely with the published values, although a handful differ by more than 5 percent, for instance, the Bogotá '95 male estimate for "Started 6th grade in private" is 0.090 versus 0.082 in the paper. Importantly, the signs and substantive magnitudes remain mostly unchanged, so our core conclusion—that vouchers increase private-school entry and reduce grade repetition—stands fully supported.

These small discrepancies stem from several sources. First, the publicly released SAS data files have been updated since the paper's publication; as the authors note in their README, "Some variables have been updated, so not all results exactly match the published values." Even slight changes in variable definitions or sample inclusion can shift means and regression slopes enough to push a few cells outside a 5 percent tolerance. Second, we lack full clarity on the covariance estimator the original authors used: our code applies statsmodels' HC0 heteroskedasticity-robust standard errors, whereas the SAS scripts may

have employed HC1 or even clustered standard errors at the lottery/strata level. Those choices can subtly reweight observations and affect both point estimates and their reported uncertainty. Finally, there are inevitable tool-specific differences when moving from SAS to Python's statsmodels: factor-variable encoding, handling of missing observations, and numeric optimization tolerances can all introduce tiny shifts in regression outputs, even when the underlying data and model are nominally the same.

Despite these modest gaps, all replication errors are within the range of acceptable "noise" given the incomplete documentation of every data-processing and estimation nuance. Crucially, the direction, statistical significance, and economic interpretation of the voucher effects remain robust across both platforms.

2.4 Tables 5 and 6

Table 5: Test Results

Variable	**OLS results**	**OLS results with covariates**	**RE**	**RE with covariates**	**Sample Size**
A. All Applicants:					
Total points	0.217	0.207	–	–	282
	(0.119)	(0.110)			
Math scores	0.177	0.156	–	–	282
	(0.119)	(0.115)			
Reading scores	0.205	0.203	–	–	282
	(0.120)	(0.114)			
Writing scores	0.126	0.131	–	–	282
	(0.121)	(0.113)			
Pooled test scores	–	–	0.170	0.165	846
			(0.095)	(0.088)	
Math and reading scores	–	–	0.191	0.183	564
			(0.102)	(0.096)	
B. Female Applicants:					
Total points	0.197	0.256	–	–	145
	(0.156)	(0.137)			
Math scores	0.281	0.326	–	–	145
	(0.147)	(0.139)			
Reading scores	0.120	0.137	–	–	145
	(0.160)	(0.153)			
Math and reading scores	–	–	0.201	0.253	290
			(0.131)	(0.117)	
C. Male Applicants:					
Total points	0.204	0.157	–	–	134
	(0.184)	(0.185)			
Math scores	0.010	-0.024	–	–	134
	(0.189)	(0.196)			
Reading scores	0.276	0.221	–	–	134
	(0.184)	(0.185)			
Math and reading scores	–	–	0.143	0.087	268
			(0.159)	(0.159)	

This table presents test results analyzing the relationship between voucher status and various academic performance indicators for different student groups. The study examines outcomes for all applicants as well as separate analyses by gender (female and male applicants). The table shows statistical results such as: OLS (Ordinary Least Squares) regression both with and without covariates, and RE (Random Effects) models with and without covariates. The types of scoring metrics are total points, math scores, reading scores, writing scores, pooled test scores, and combined math and reading scores.

The numbers in each cell represent coefficient estimates, with standard errors shown in parentheses below each estimate. The sample sizes for each analysis are provided in the rightmost column. According to the notes, the standard errors are robust and corrected for within-school-of-application clustering. The covariates include controls for applicant's age, gender, parents' schooling, strata of residence, and type of survey and instrument.

Overall, this table explores whether receiving a voucher has significant effects on standardized test performance, examining these effects across different subject areas and for different demographic groups.

2.4.1 Challenges

When attempting to decipher the SAS file, there was difficulty when trying to filter for the covariates for math and reading as well as the pooled test scores. The code for math and reading and pooled test scores with covariates include all of the covariates the paper mentions and the SAS file used, however the numbers are not accurate and this might be due to missing data or issues with the data itself. However, these columns are random effects models which we hadn't learned in class yet, but I still wanted to attempt.

Additionally, when filtering for female and male data, the coefficients were very inaccurate and not much explanation as to why that is. Even after consulting course staff for help, there might have been filtering on the data that was not mentioned in the SAS file or paper that came to that final regression in the article. After trial and error, I used a different variable

than what was used in the SAS file suggested instead of sex because the original variable used led to inaccurate coefficients. There was still issue with including the covariates and after trying to read through the paper and deciphering the SAS file, I concluded that the covariates that were used are the same as the covariates used for the all applicants panel. The paper also mentioned that the table used robust standard error, however after attempting to use robust standard error, it was not very accurate for the standard error.

Table 6: Non-Educational Outcomes and Voucher Status

Panel	Dependent variable	Bogotá 1995		Combined sample	
		Loser means	Basic Controls	Loser means	Basic Controls
A. Male and Female:					
	Married or living with companion	0.0160 (0.1256)	-0.0087 (0.0062)	0.0171 (0.1297)	-0.0091 (0.0055)
	Has child	0.0338 (0.1809)	-0.0103 (0.0096)	0.0303 (0.1714)	-0.0064 (0.0078)
	Applicant is working	0.1690 (0.3751)	-0.0296 (0.0204)	0.1618 (0.3685)	-0.0266 (0.0170)
	Number of hours working	4.8808 (12.3107)	-1.2189 (0.6365)	4.4224 (11.6027)	-0.8724 (0.5197)
	Sample size	562	1,146	760	1,575
B. Male:					
	Married or living with companion	0.0036 (0.0598)	-0.0039 (0.0036)	0.0027 (0.0518)	-0.0027 (0.0026)
	Applicant is working	0.2321 (0.4230)	-0.0366 (0.0334)	0.2258 (0.4187)	-0.0298 (0.0282)
	Number of hours working	6.4214 (13.6920)	-0.6376 (1.0684)	6.2151 (13.3276)	-0.5268 (0.8938)
	Sample size	280	575	372	778
C. Female:					
	Married or living with companion	0.0284 (0.1663)	-0.0099 (0.0118)	0.0309 (0.1733)	-0.0027 (0.0026)
	Pregnant or has child	0.0603 (0.2384)	-0.0151 (0.0179)	0.0541 (0.2266)	-0.0092 (0.0144)
	Applicant is working	0.1064 (0.3089)	-0.0312 (0.0236)	0.1005 (0.3011)	-0.0266 (0.0170)
	Number of hours working	4.8808 (12.3107)	-1.2189 (0.6365)	4.4224 (11.6027)	-0.0298 (0.0282)
	Sample size	282	571	388	778

Table 6 compares voucher winners and losers in male, female, and combination populations, focusing on noneducational outcomes such as marriage, childbirth, participation in the labor market, and weekly working hours. The coding estimated the treatment effects of earning a voucher on various life outcomes using reduced-form OLS models with simple

controls. This table is important because it tests for wider societal effects that can result from greater educational possibilities, taking the study beyond educational measurements. Standard errors are presented to illustrate accuracy; these estimates use group-fixed effects to control for potential confounders, making the results interpretable within a framework of causal inference. The results support the paper's overarching argument that the PACES vouchers affected life paths in addition to education, lowering early labor force entry and postponing the start of families.

2.4.2 Challenges

For Table 6, the regression results align closely with those reported in the original article, with some minor coefficient variations, which fall within the acceptable range outlined by the course rubric (e.g., within ± 5 percentage of the original estimates).

The paper also refers to "Basic Controls" in the regression table, and we had to infer based on the SAS file that is publicly available to recreate the table. There was difficulty trying to identify these controls due to the lack of information about the table columns and the article not being specific enough. We concluded that the controls used were age, phone access, sex, interview month, and socioeconomic group.

The original Table 6 also included a column about Probit with basic controls, but we have not yet learned that concept, which is why we chose to exclude it.

2.5 Tables 7, 8, 9

Table 7 estimates the causal impact of earning a private school scholarship on various educational and non-educational outcomes using a Two-Stage Least Squares (2SLS) regression:

First stage regression:

$$y_i = \mathbf{X}_i' \boldsymbol{\beta}_1 + \alpha_1 s_i + \xi_i \tag{2}$$

Second stage regression:

$$s_i = \mathbf{X}_i'\boldsymbol{\gamma} + \pi Z_i + \eta_i \tag{3}$$

Using lottery wins as an instrument to account for non-random scholarship uptake, the table compares results between the Bogotá 1995 subsample as well as the entire combined sample. Highest grade completed, grade repetition, enrollment status, and standardized test scores are among the key dependent variables. The results of the2SLS regression show larger estimates compared to their OLS counterparts for educational outcomes, which illustrates how strong the causal impact of vouchers is. This table was created through regression modelling with robust standard errors and instrumental variable procedures, and its estimation emphasizes the importance of using randomized voucher assignment to isolate the true causal effect of scholarship use.

The 2SLS results consistently show larger estimates than their OLS counterparts for educational outcomes, such as a 0.13 to 0.20 year increase in grade attainment and a 13-15 percent increase in 8th grade completion. This demonstrates the strength of the voucher's causal impact, and the effect on test scores shows the program's learning gains. This table was constructed through regression modeling using robust standard errors and instrumental variable techniques, and its estimates emphasize the importance of using randomized voucher assignment to isolate the true effect of scholarship use.

Table 8 estimates the impact of winning a voucher on school enrollment and tuition-related outcomes for Bogotá 1995 applicants, comparing between the full sample and specifically private school students. The table reports these outcomes with and without basic controls, like phone access, age, survey type and instrument, strata of residence, and month of interview.

Table 9 reports the upper and lower bounds of the voucher's effect on matriculation and tuition fees for Bogotá 1995 applicants who would have attended private school – even without the voucher – but paid more than they received from the voucher. Voucher winners within the upper half of the fee distribution spent around 38-48 dollars more on private

school fees than they would have paid, if not for the voucher.

2.5.1 Challenges

Despite extensive effort, we were unable to fully reproduce Table 7 from Angrist et al.(2002) using the data and code provided. While we had access to the tab7.dta and tab7test.dta datasets, the associated SAS replication code (table7_final.sas), and the published paper, several issues prevented us from replicating the table exactly:

The SAS code includes a set of control variables referred to as "basic controls", it does not document how each variable name corresponds to the specific control, which makes deciphering the included controls difficult. Additionally, any variable transformations (like squared terms and dummy encoding for categorical variables) are embedded in the code but not mentioned in any comments, making it difficult to verify whether our reconstructed models match those used in the original analysis.

Another major challenge in attempting to reproduce Table 7 was that the control variables required for the regressions appear to originate from different datasets, tab7.dta and tab7test.dta. When we attempted to construct the full set of control variables described in the paper, we found that many of these variables had non-overlapping missing values. As a result, there was a discrepancy in the number of observations between the instrument, the endogenous regressor, and the control matrices. Attempting to run a 2SLS regression with these mismatched matrix sizes led to errors that prevented us from replicating Table 7.

Tables 8 and 9 show the effect of winning a voucher on school matriculation and tuition among Bogotá 1995 applicants. In the paper, the notes beneath Table 8 describe that sample sizes differ across rows due to missing fee data. This same fee data is used in Table 9, to create the dollar cutoffs across which the voucher's effect is measured. This means that both Table 8 and 9 are composed of inaccessible fee data and, as a result, are not replicable.

Further complicating the issue, Tables 8 and 9 are the only tables without specified SAS files. The readme.txt describes which SAS file can be used to create each of the paper's

tables. Table 8 and Table 9 are the only exceptions to this and, as a result, are not fully replicable.

3 Conclusion

This replication of the paper *Vouchers for Private Schooling in Colombia: Evidence from a Randomized Natural Experiment* reinforces the original paper's findings on the impact of the PACES program, which issued vouchers to partially cover the cost of private secondary schooling. The vouchers had a significant and positive effect on the educational outcomes of voucher recipients by increasing private school attendance, decreasing grade repetitions, and reducing student involvement in the labor force.

The most significant results of our replications would be Tables 3, 5, and 7, as they provided the strongest evidence for the effectiveness of PACES vouchers in improving educational outcomes. Table 3 reports the increased probability of voucher recipients attending private school, in comparison to those who did not win the vouchers. Table 5 indicates that voucher recipients scored higher on achievement tests, which were issued three years after voucher winners received the PACES vouchers. Table 7 quantifies the vouchers' effect on reducing total grade repetitions and increasing probabilities of completing eighth grade among voucher recipients. These tables directly support the paper's findings that the vouchers have positive impacts on the educational outcomes, shown by the increased probabilities of affording private schooling, grade completion, and higher performance on achievement tests. These are clear indicators that the voucher program had a positive impact on a range of educational outcomes, ranging from school type to actual test scores.

This replication project not only supports the original findings, but also highlights the importance of randomized controlled trials in evaluating educational policies. By distributing PACES vouchers through a lottery, the original study was able to isolate the causal impact of private school access from confounding factors. This is shown in Table 2, which measured

the effect of personal characteristics – such as telephone access, age, and sex – on one's chances of winning a voucher.

Randomized controlled trials offer reliable evidence by isolating causal effects from confounding variables, which allows for more informed policy decisions. This paper's findings shows that demand-side policies, like offering vouchers to low-income students, can significantly improve educational outcomes even beyond "the [24 dollar] per winner additional cost to the government of supplying vouchers instead of public-school places" (Angrist et al. 2002, 1535). This paper asserts that the PACES voucher program reaps benefits worth more than its initial government costs, suggesting that expanding similar programs would be a less costly – yet efficient – method of reducing inequality in the long-term.

Through this project, we gained a better understanding of replicating randomized assignment studies, data cleaning and analysis, and turning empirical analyses into effective economic policies.

4 References

Joshua D. Angrist et al. 2002. "Vouchers for Private Schooling in Colombia: Evidence from a Randomized Natural Experiment ." The American Economic Review, 92(5), 1535-1558